Camphene to Camphor

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N/A

11/28//23

Chem 2081-003

Introduction

The purpose of this experiment was to take borneol and convert to camphor. Borneol is an alcohol product that is helped to prepare other products. It can be found in fragrances mostly around households. In medicine it is used as a transition drug or messenger drugs. It is mostly used for these purposes as a sum of the parts leading to the whole. Camphor is an active ingredient in many medicines such as itch creams and chest decongestants.

Materials and Methods

To start the lab a sand bath was prepped. Then, in a long neck round bottom flask .3 grams of boreneol was added from day 2 along with .3 mL of glacial acetic acid then the racemic mixture was dissolved in acetone. A reflux column was assembled, and the solution was allowed to fully dissolve the boreneol and was refluxed for about 30 minutes. After dissolving 6 mL of bleach was added dropwise after the last bit of bleach was added the reaction continued for about 15 minutes. Next, the solution was allowed to cool then washed with methylene chloride in a centrifuge tube for easier extraction. Then the solution was neutralized using sodium bicarbonate. Then the organic product was washed with sodium bisulfite and 2 mL of water. The product was dried with sodium sulfate and then the DCM was evaporated over a water bath revealing the product crystals.

Results

Weight of Dry Watch Glass (g)	37.322

Weight of Product + Watch Glass (g)	37.496
Weight of Product (g)	.174

Table 1. Weight of watch glass, Product + watch glass, and product

Calculations

On lab Page.

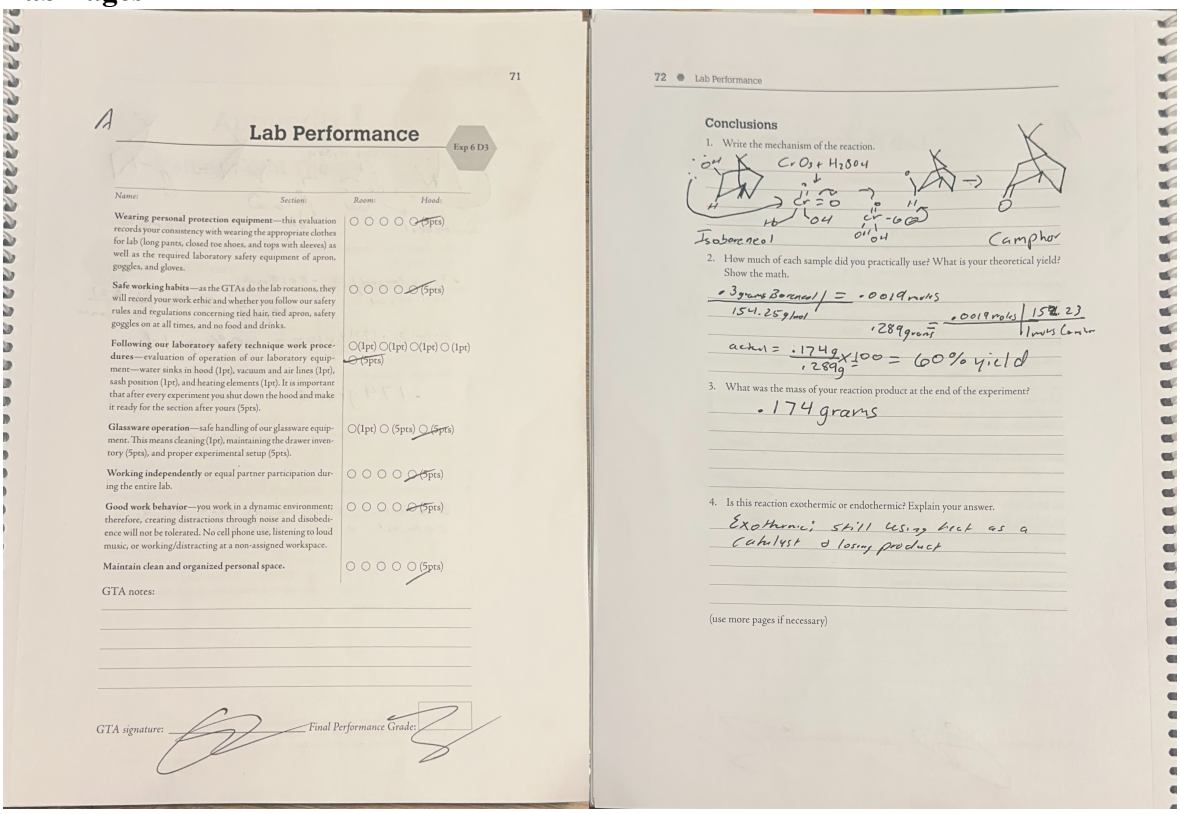
Discussion

In this experiment borneol was taken from day 2 and reacted to yield the final product of camphor. Borneol is an important intermediate and is used often. The yield was around 60% yield with a couple of sources of error and observations noticed. For one, the reactant borneol was made in lab as a product from day 1 plus reactants. If the borneol was not pure and had imperfections it could affect the percent yield by allowing side products. The reflux is an irreversible reaction and should have been pretty accurate if allowed to go all the way through so these possible sources of error. If I could change anything I would make sure the reflux happens completely by some sort of understanding of boiling point and not based on time. This would be the only thing that could be changed at this point in the experiment.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.

Lab Pages



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A

Lab Performance

Exp 6 D3

Name: Section: Room: Hood:

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. ○ ○ ○ ○ ○ (5pts)

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. ○ ○ ○ ○ ○ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). ○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). ○(1pt) ○ (5pts) ○ (5pts)

Working independently or equal partner participation during the entire lab. ○ ○ ○ ○ ○ (5pts)

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. ○ ○ ○ ○ ○ (5pts)

Maintain clean and organized personal space. ○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: Final Performance Grade:

72 ● Lab Performance

Conclusions

1. Write the mechanism of the reaction.

Isoborneol Camphor

2. How much of each sample did you practically use? What is your theoretical yield? Show the math.

.3 grams Borneol / 154.25 g/mol = .0019 moles

.0019 moles | 1 | 152.23 / 1 moles Camphor = .1289 grams

actual = .174g / .1289g × 100 = 60% yield

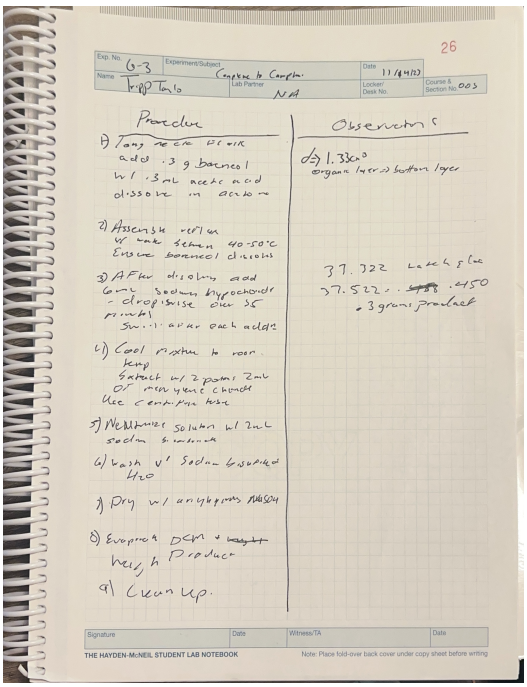
3. What was the mass of your reaction product at the end of the experiment?

.174 grams

4. Is this reaction exothermic or endothermic? Explain your answer.

Exothermic; still using heat as a catalyst & losing product

(use more pages if necessary)



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Exp. No. G-3 | Experiment/Subject Complete & Complete | Date 11/14/22

Name Trip Taylo | Lab Partner NA | Locker/Desk No. | Course & Section No. 003

Procedure

1) Long reflux of crude acid .3 g borneol w/ .3 mL acetic acid dissolve in acetone

2) Assemble reflux w/ water bath 40-50°C Ensure borneol dissolves

3) After dissolving add conc. sodium hypochlorite - dropwise over 35 minutes Swirl after each addition

4) Cool mixture to room temp Saturate w/ 2 pams 2mL of methylene chloride Use centrifuge tube

5) Neutralize solution w/ 2mL sodium bicarbonate

6) Wash w/ Sodium bicarbonate H_2O

7) Dry w/ anhydrous $MgSO_4$

8) Evaporate DCM & weigh Product

9) Clean up.

Observations

a) 1.33cm³ organic layer → bottom layer

37.722 Lost flow

37.522 - .4450

.3 grams product

Signature | Date | Witness/TA | Date

THE HAYDEN-McNEIL STUDENT LAB NOTEBOOK Note: Place fold-over back cover under copy sheet before writing